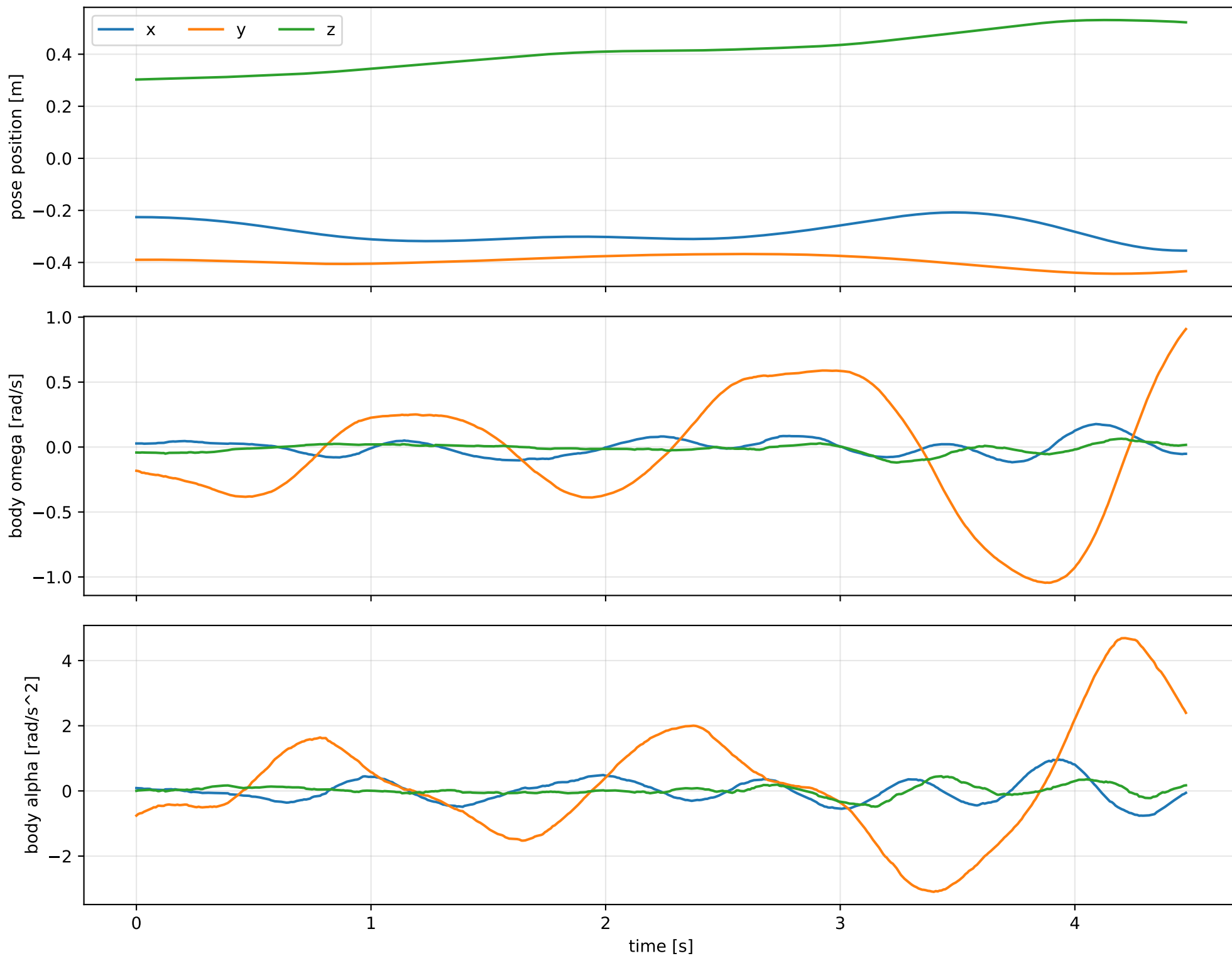
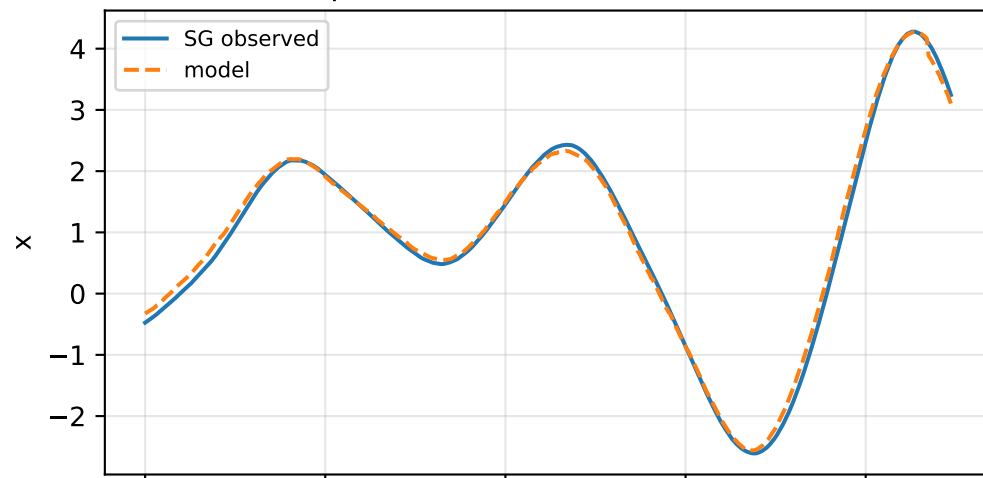
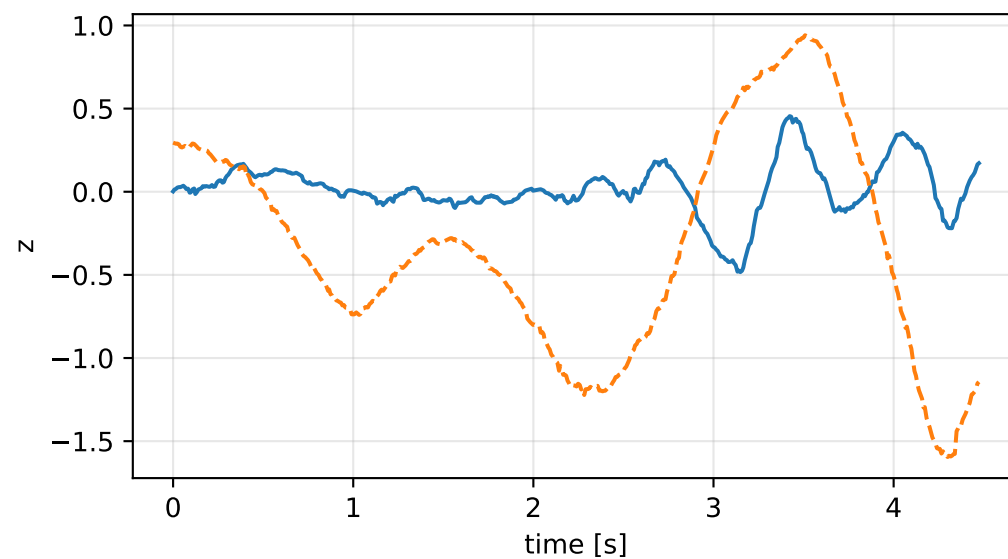
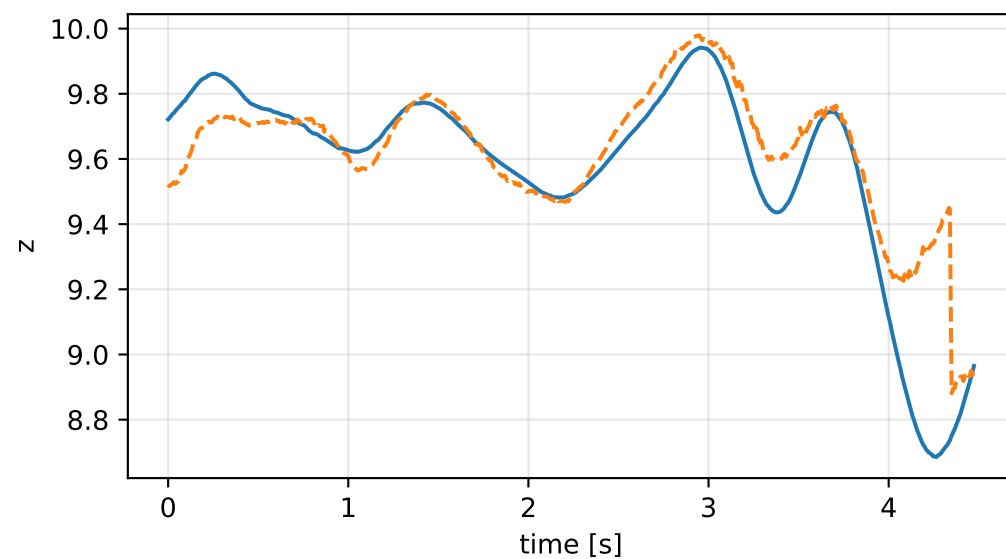
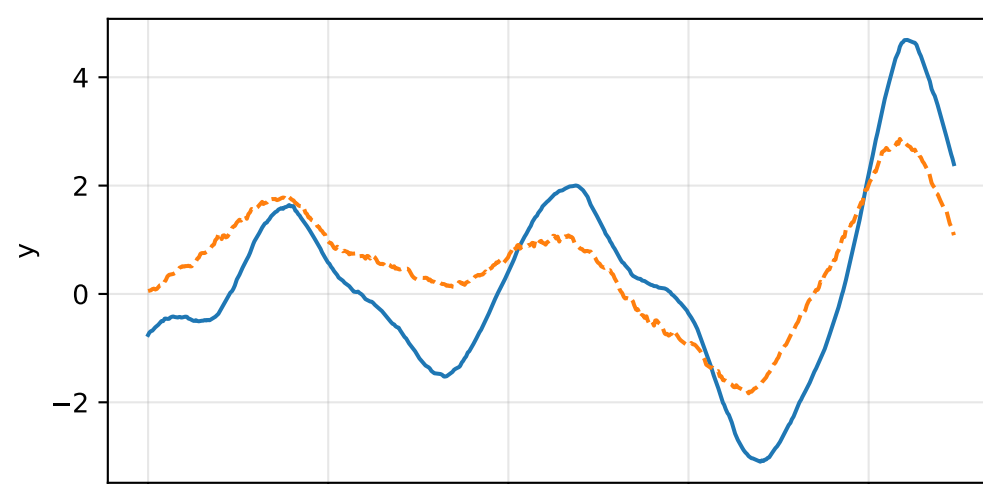
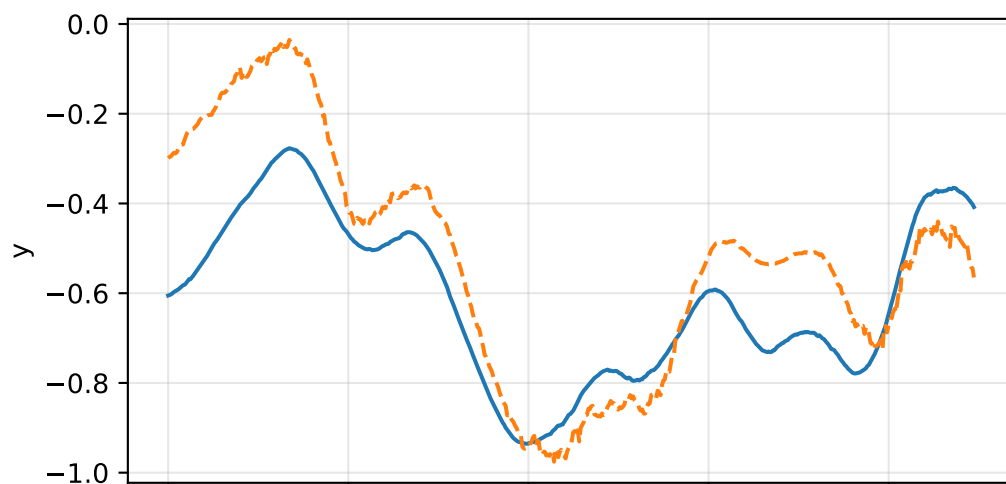
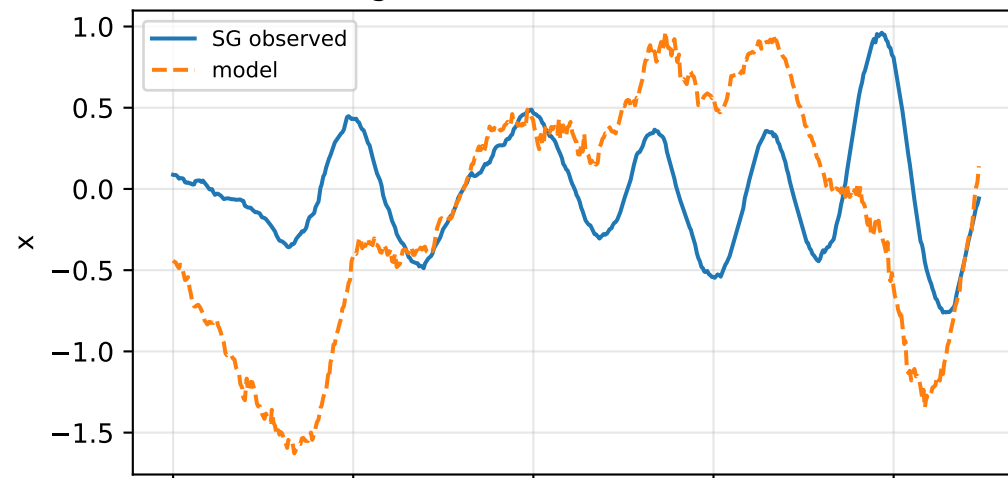


cov_full_no_position_rotation_cross: SG trajectory and derived motion

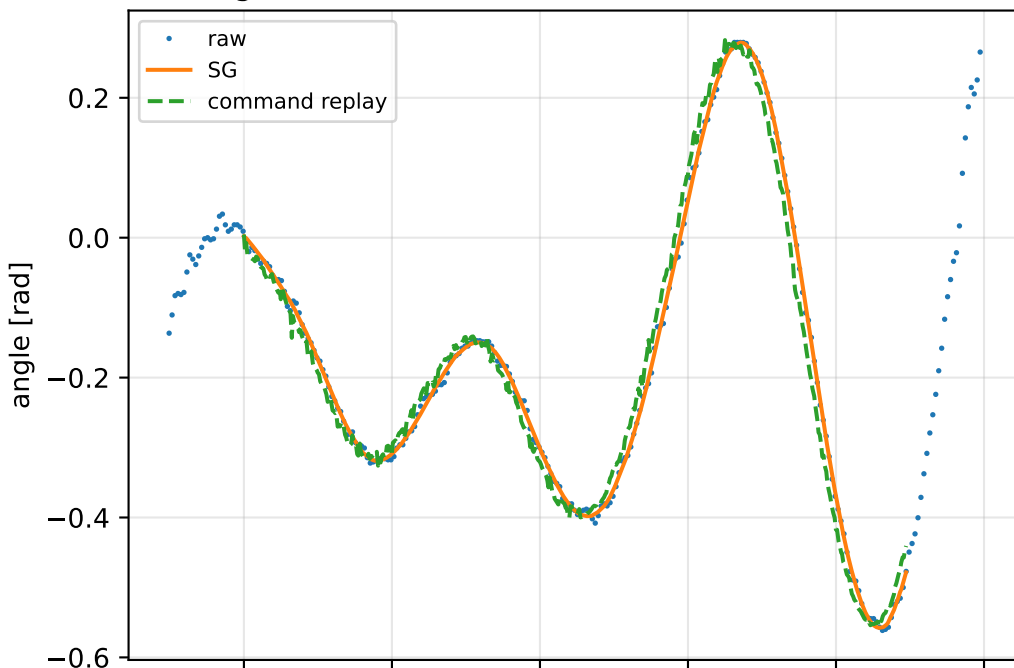


cov_full_no_position_rotation_cross: acceleration residual objective

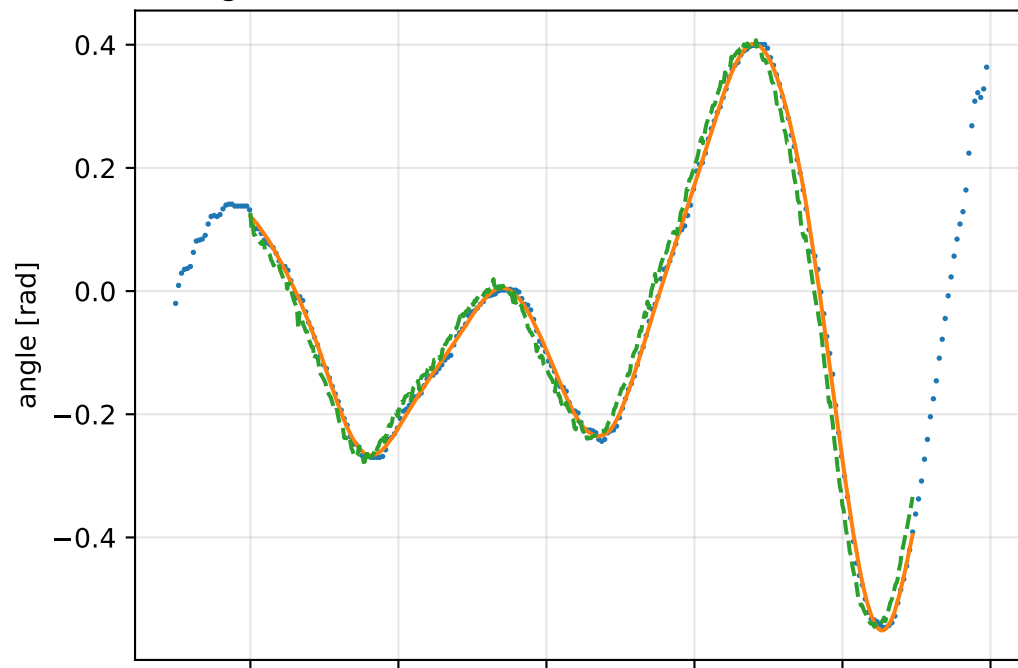
specific acceleration [m/s²]angular acceleration [rad/s²]

cov_full_no_position_rotation_cross: gimbal measurement audit (objective=measured_sg)

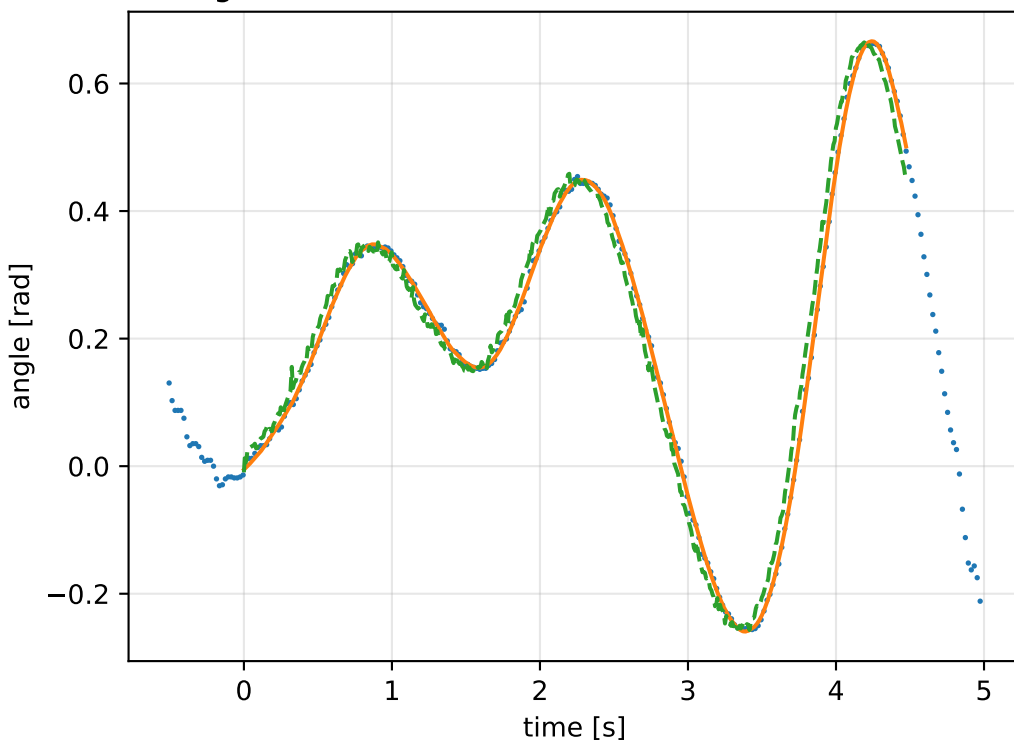
gimbal 1: RMSE=0.027, max=0.07529 rad



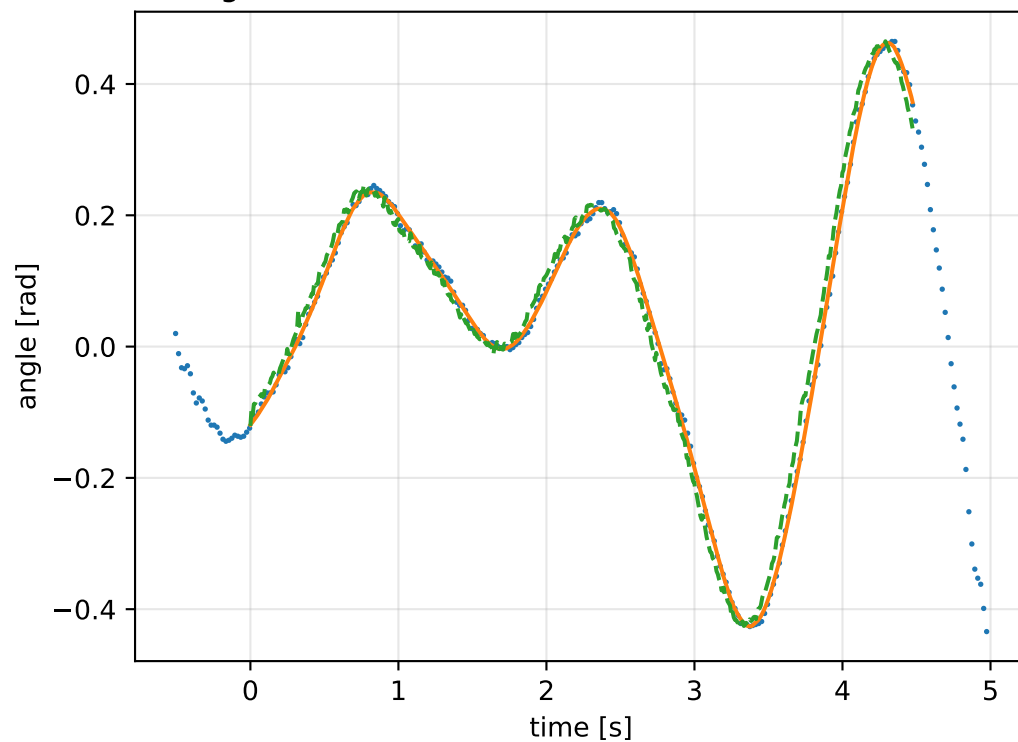
gimbal 2: RMSE=0.03107, max=0.08239 rad



gimbal 3: RMSE=0.03106, max=0.08682 rad

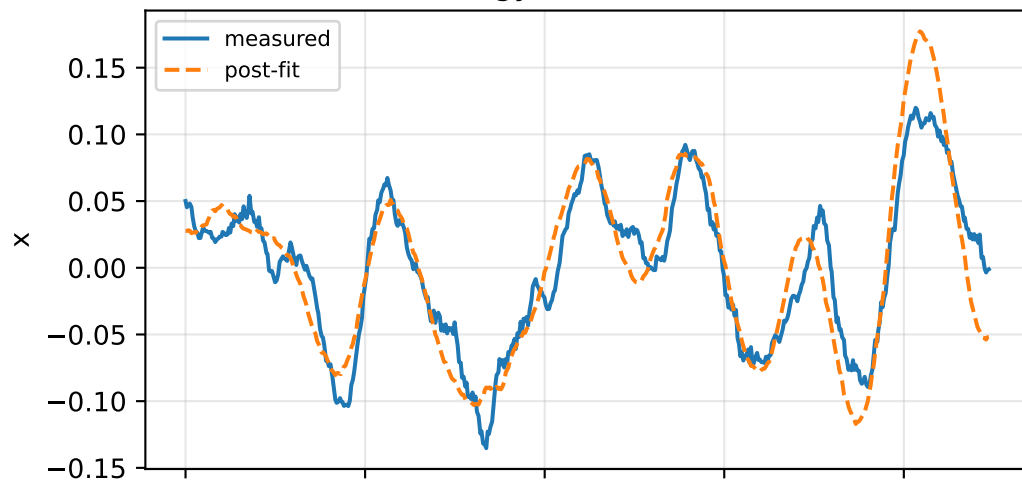


gimbal 4: RMSE=0.02598, max=0.07261 rad

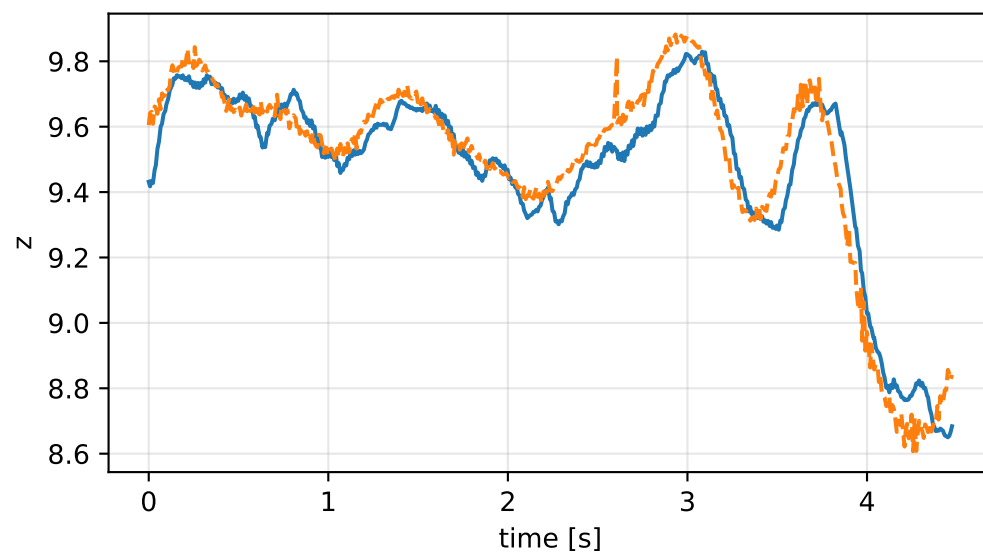
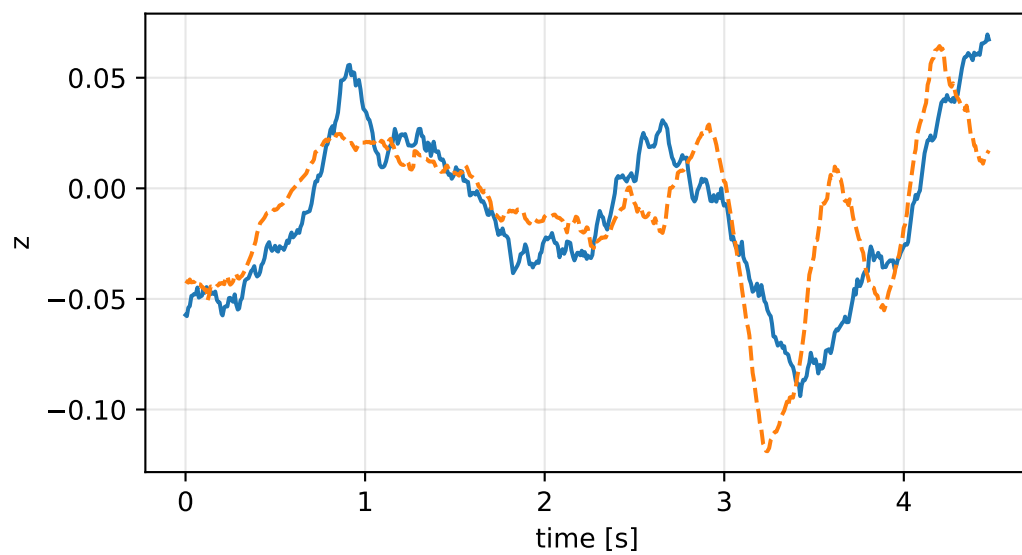
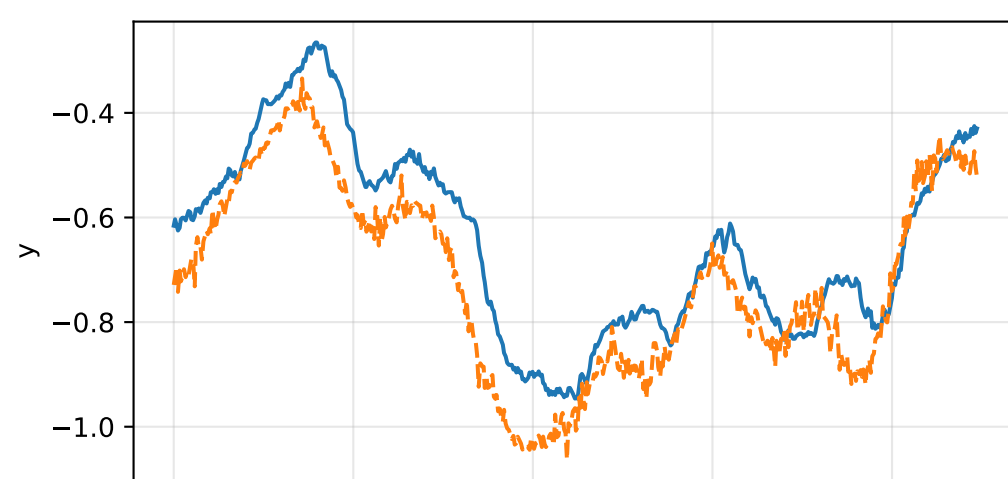
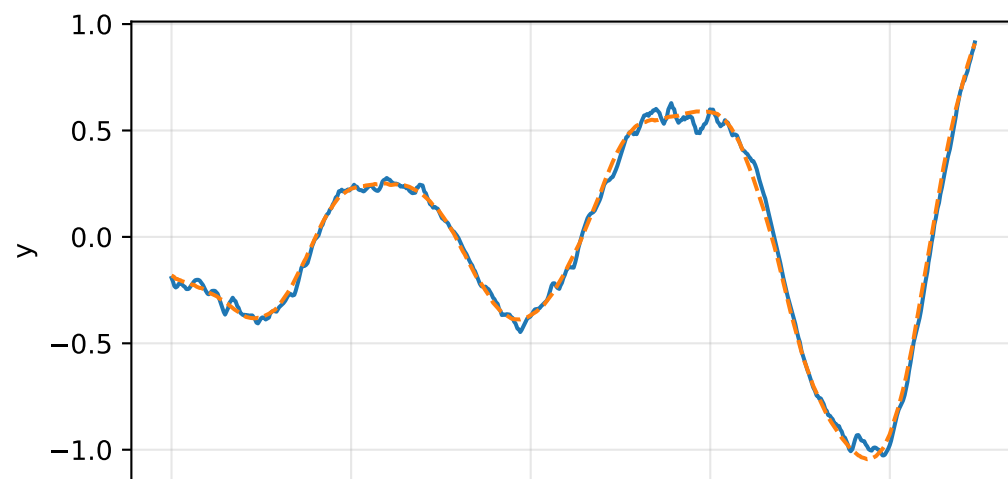
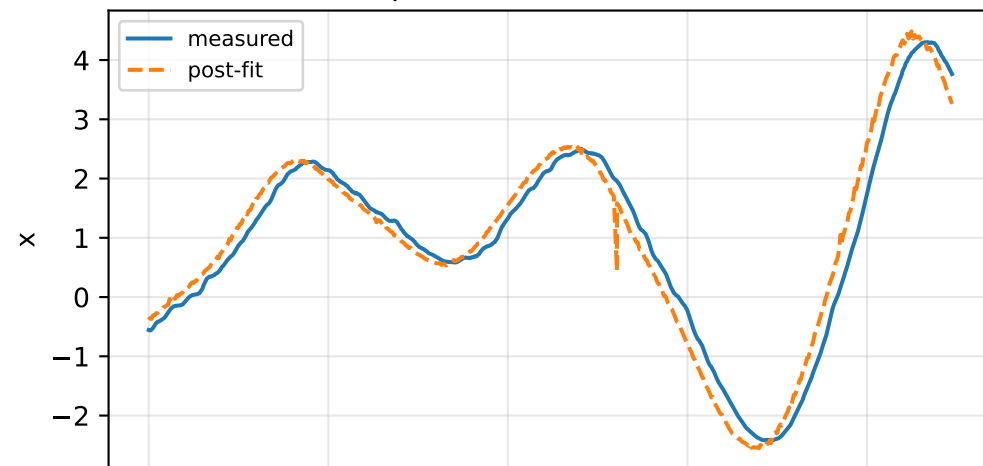


cov_full_no_position_rotation_cross: IMU comparison (diagnostic only)

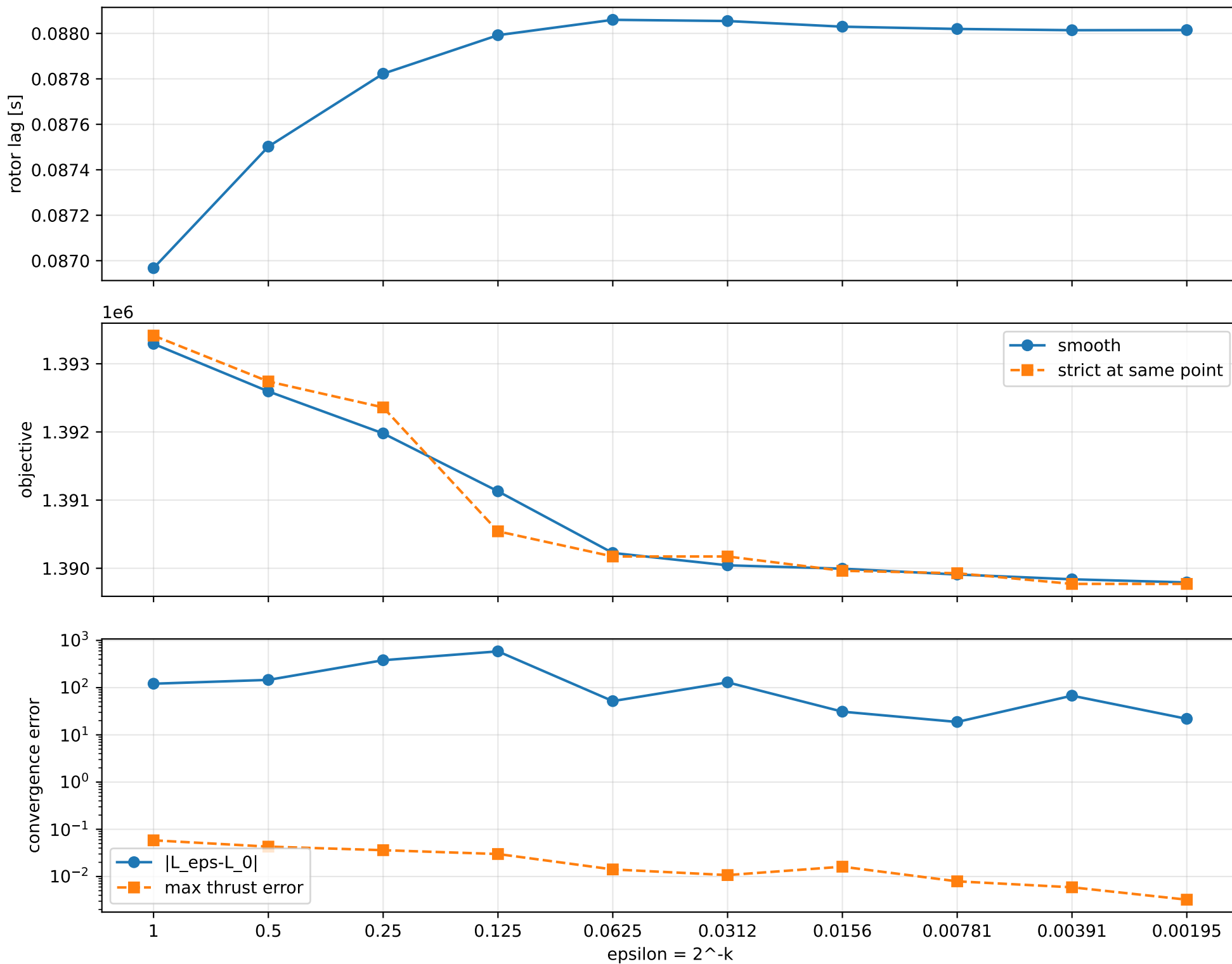
gyro [rad/s]



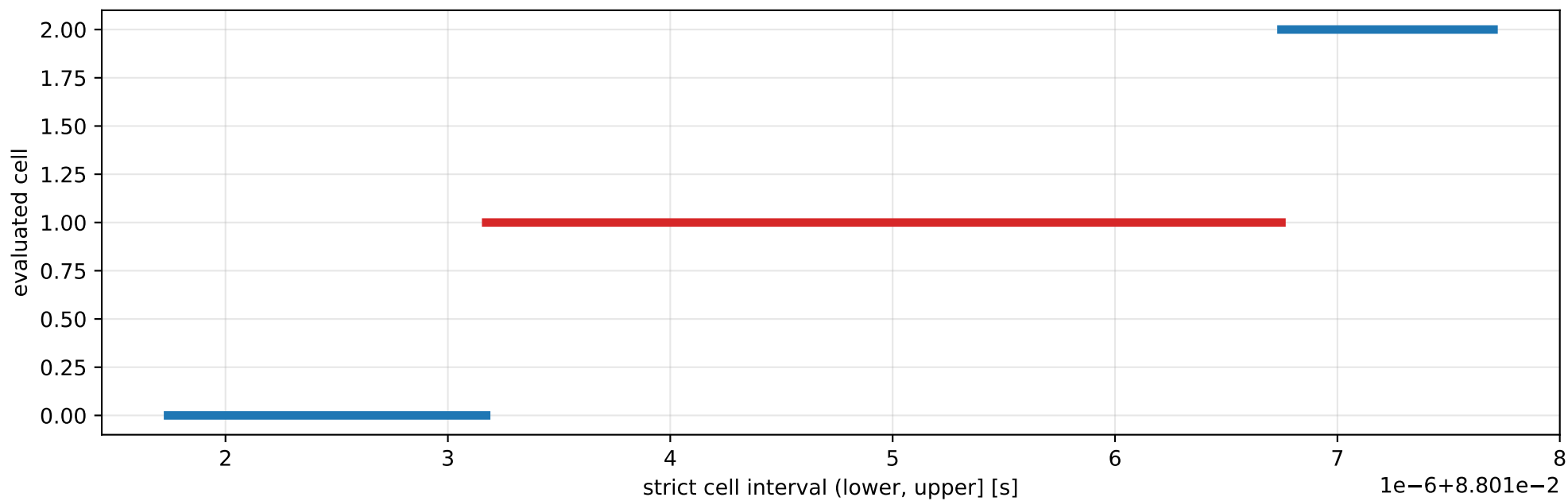
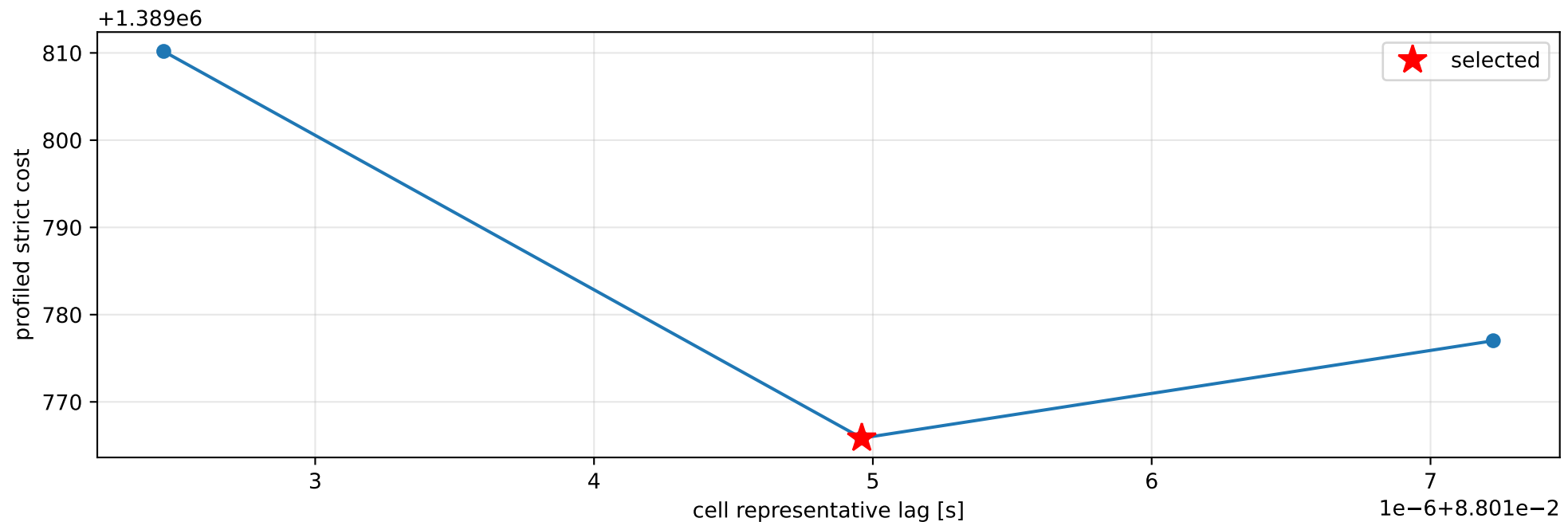
specific force [m/s^2]



cov_full_no_position_rotation_cross: smooth-to-strict continuation



cov_full_no_position_rotation_cross: exact strict-ZOH lag cells



selected (0.0880131721, 0.0880167484] s; final smooth 0.0880146871 s; data boundary=False

cov_full_no_position_rotation_cross: parameter estimate

Case: cov_full_no_position_rotation_cross
status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 15.5768932

inertia [kg m²]:

```
[[ 1.8119588  1.74814421  0.92900204]
 [ 1.74814421  3.88126677 -0.77007403]
 [ 0.92900204 -0.77007403  4.8307966  ]]
```

CoG body [m]: [0.06434486 0.18172351 -0.11642656]

force effectiveness: [4.43766870e-05 2.53826025e+00 1.22005474e+01 7.83189907e+00]

scale-free J/m [m²]:

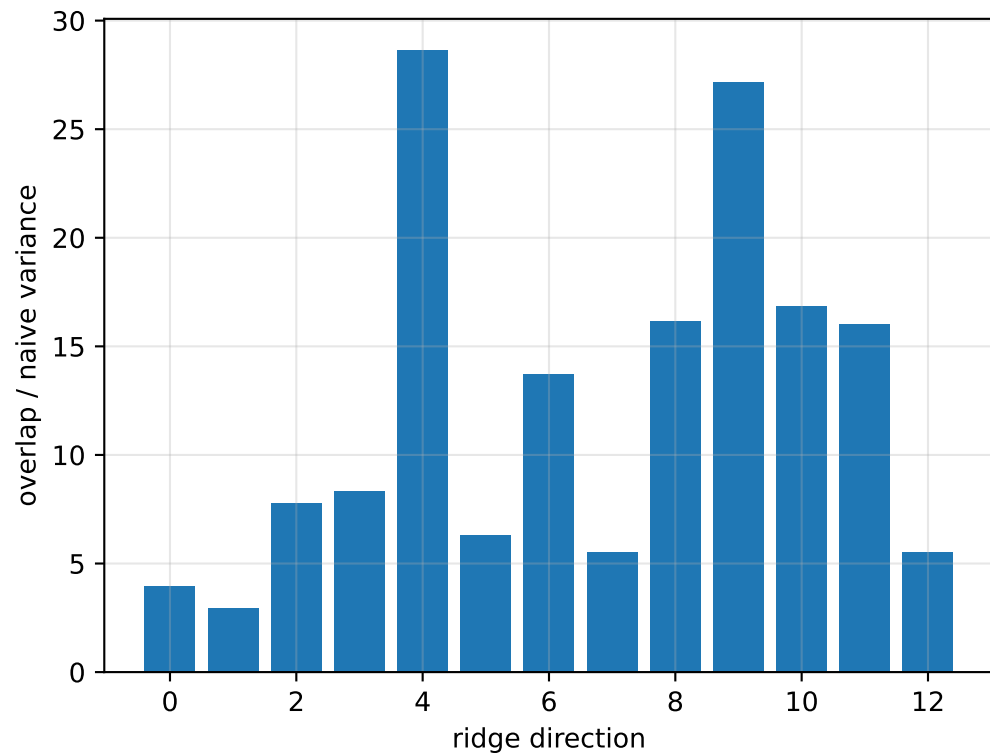
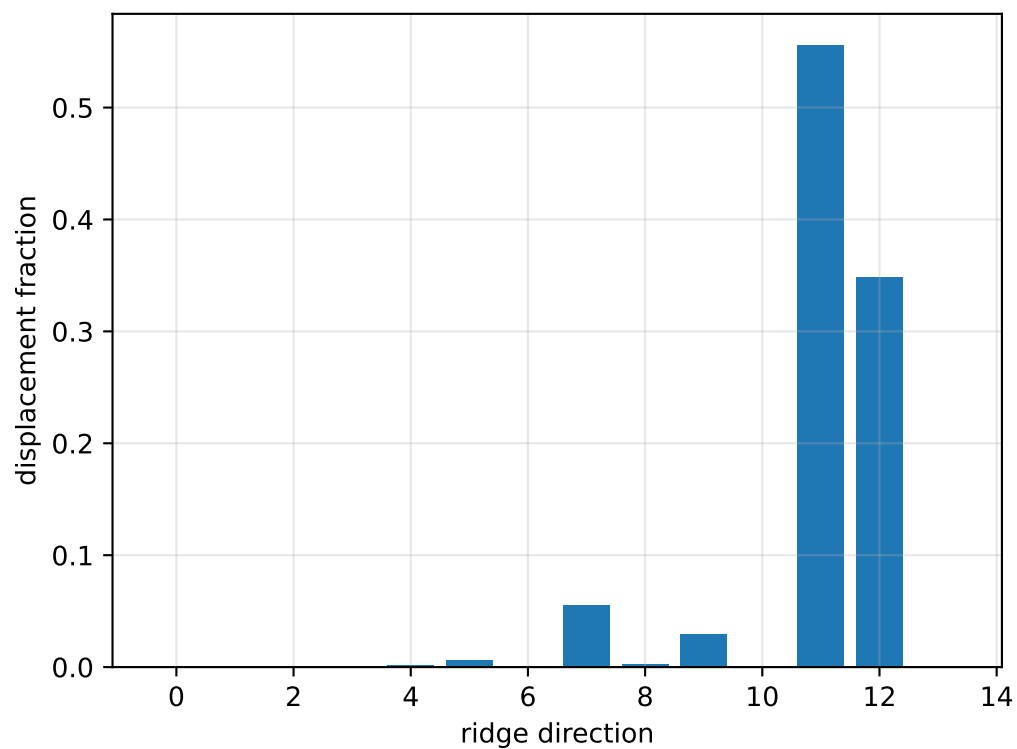
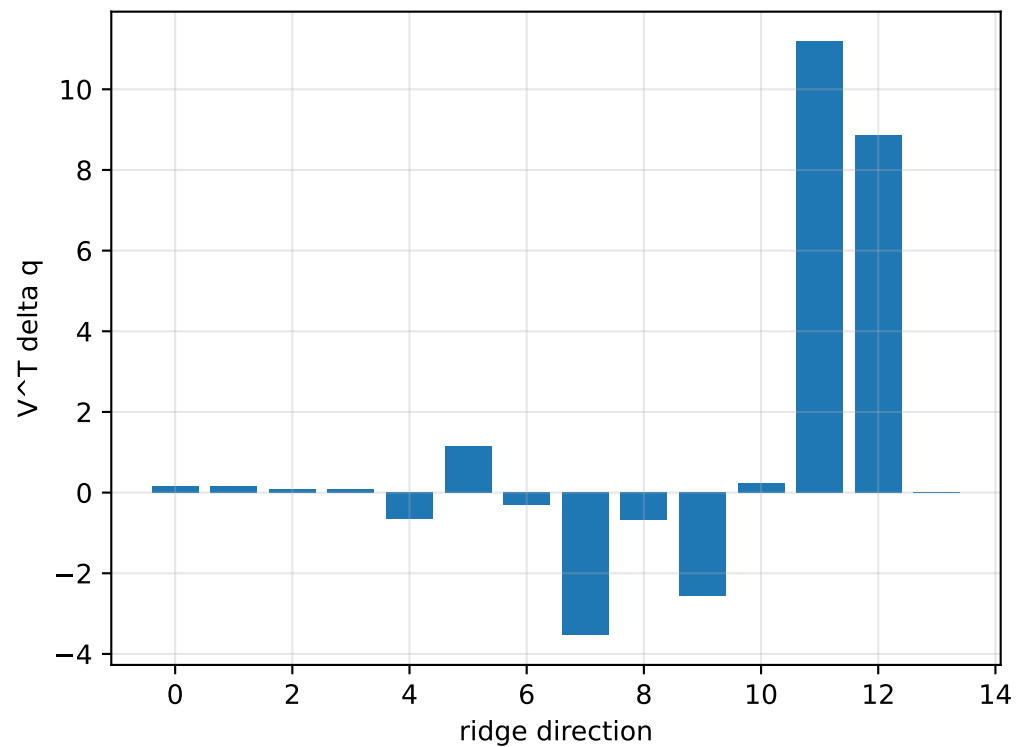
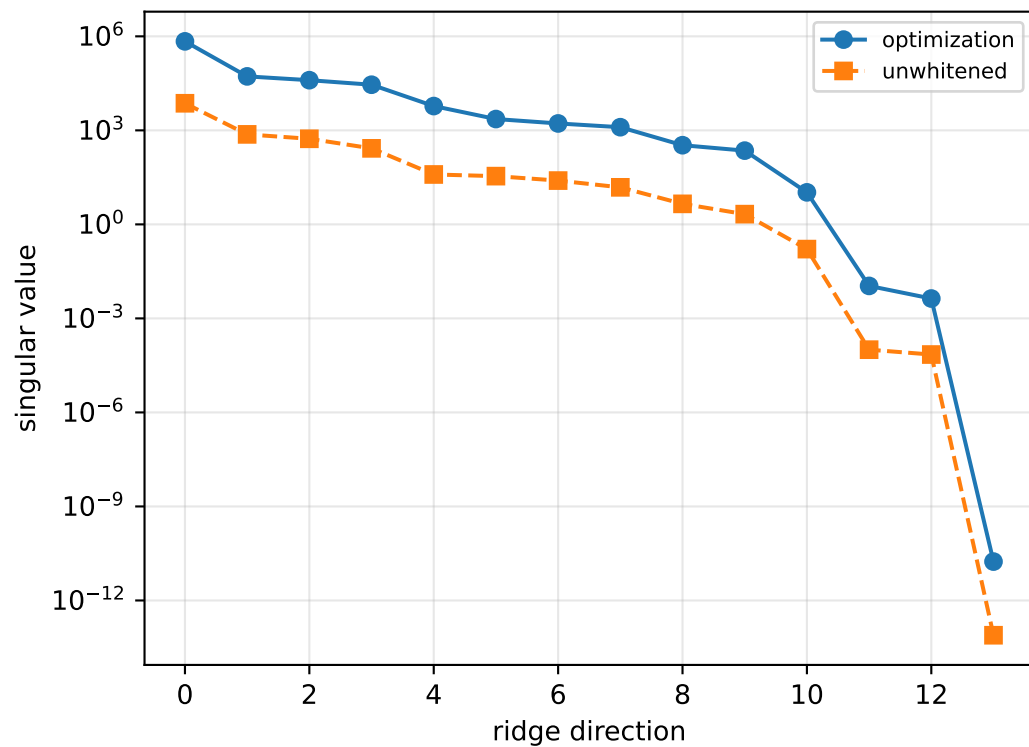
```
[[ 0.1163235  0.11222676  0.05963975]
 [ 0.11222676  0.24916822 -0.04943695]
 [ 0.05963975 -0.04943695  0.31012581]]
```

scale-free f/m: [2.84887919e-06 1.62950353e-01 7.83246521e-01 5.02789545e-01]

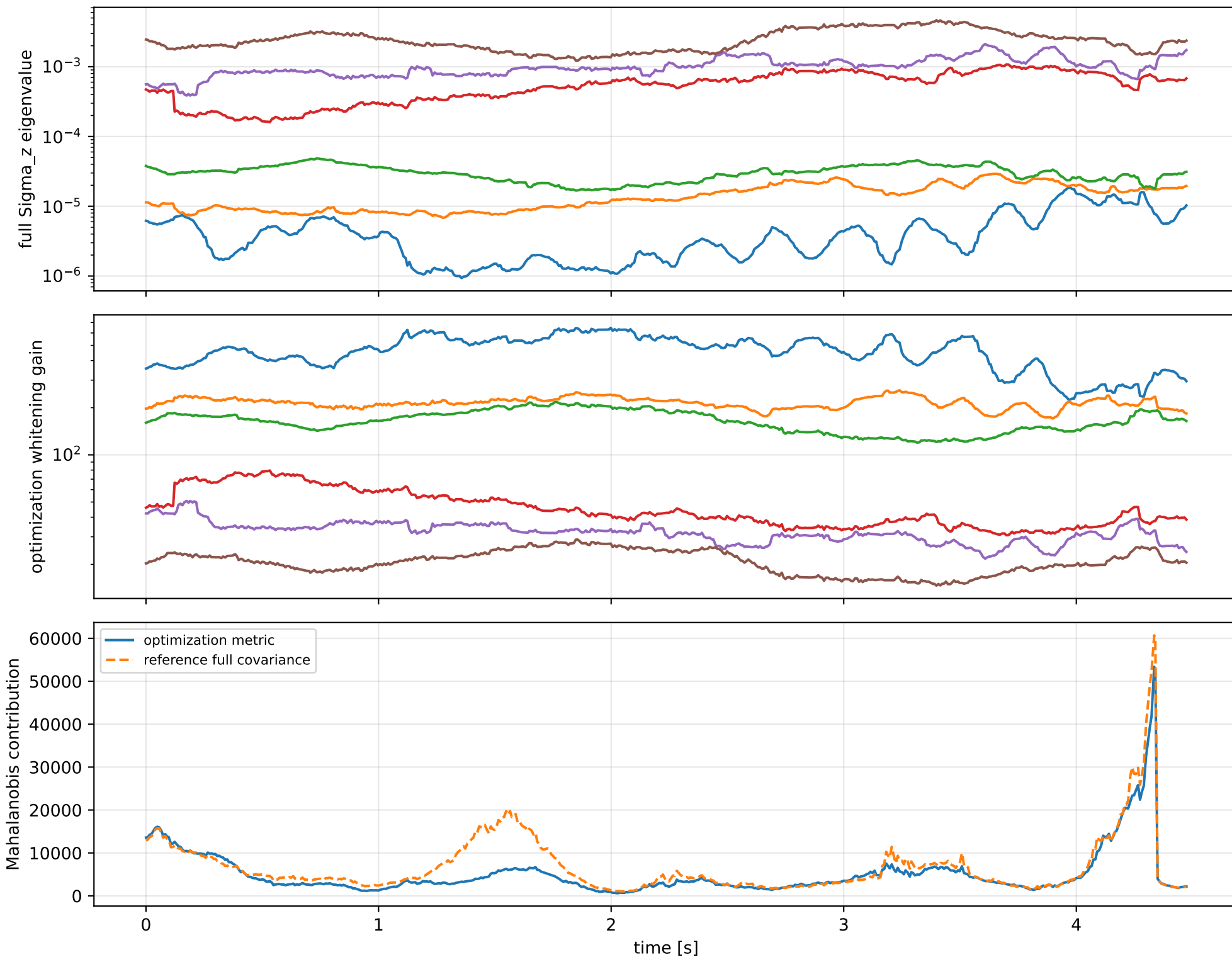
rotor lag cell: (0.0880131721, 0.0880167484) s

representative: 0.0880149603 s

cov_full_no_position_rotation_cross: ridge spectrum (rank 13, nullity 1, ||Jv||=2.06e-11)



cov_full_no_position_rotation_cross: optimization vs reference covariance



cov_full_no_position_rotation_cross: wrench / closure (force max $4.97\text{e-}14$, torque max $3.55\text{e-}15$)

